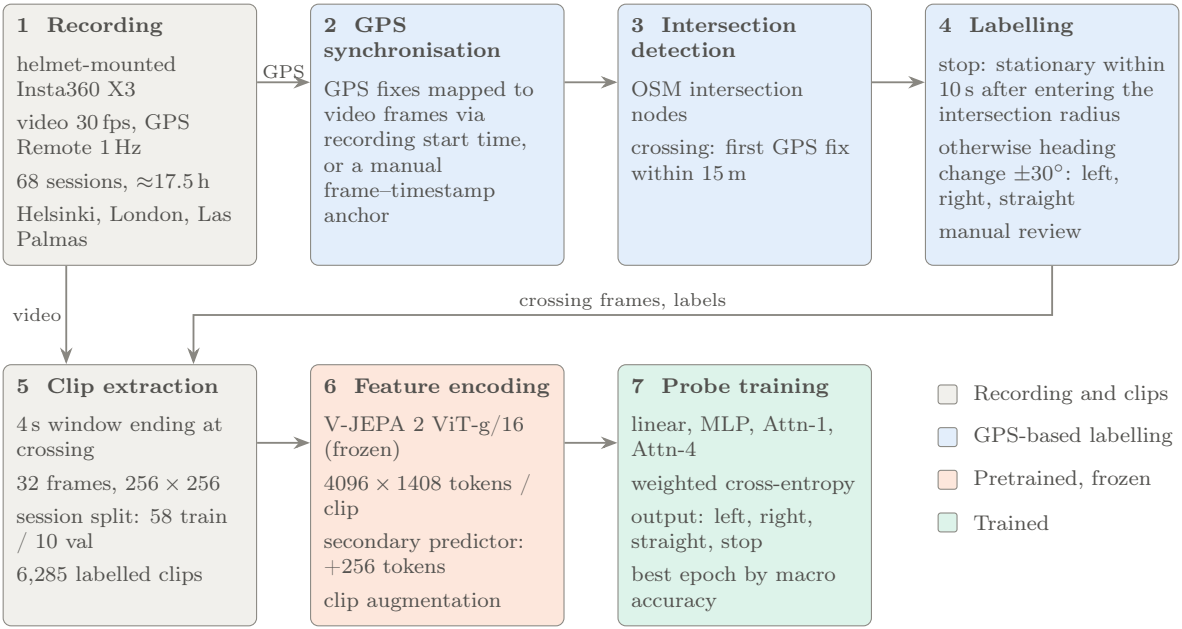

1 Recording
helmet-mounted Insta360 X3
video 30 fps, GPS Remote 1 Hz
68 sessions, ≈17.5 h
Helsinki, London, Las Palmas
GPS
2 GPS synchronisation
GPS fixes mapped to video frames via recording start time, or a manual frame–timestamp anchor
3 Intersection detection
OSM intersection nodes
crossing: first GPS fix within 15 m
4 Labelling
stop: stationary within 10 s after entering the intersection radius
otherwise heading change ±30°: left, right, straight
manual review
crossing frames, labels
video
5 Clip extraction
4 s window ending at crossing
32 frames, 256 × 256
session split: 58 train / 10 val
6,285 labelled clips
6 Feature encoding
V-JEPA 2 ViT-g/16 (frozen)
4096 × 1408 tokens / clip
secondary predictor: +256 tokens
clip augmentation
7 Probe training
linear, MLP, Attn-1, Attn-4
weighted cross-entropy
output: left, right, straight, stop
best epoch by macro accuracy
Recording and clips
GPS-based labelling
Pretrained, frozen
Trained